

Prime-power congruences for compositional tree series

A counterexample, exact modular corrections,
and a finite-core theorem for an OEIS family

For the unique formal series

$$A_\ell(x) = x \exp(A_\ell^{\circ \ell}(x)), \quad A_\ell(x) = \sum_{n \geq 1} a_\ell(n) \frac{x^n}{n!},$$

the conjecture $a_5(n) \equiv n \pmod{5}$ is false. Its first failure is $n = 23$, and all failures are precisely

$$n = 20j + 3 \quad \text{or} \quad n = 20j + 4, \quad j \geq 1.$$

The mechanism is a general prime-power congruence for labelled trees with integer types. It also proves eleven other conjectural assertions in the four OEIS entries considered here.

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Self-contained mathematical proofs; exact-arithmetic verification;
LaTeX source, Python programs, and machine-readable certificates.

Abstract

We study the integer coefficients of the formal functional equation $A_\ell = x \exp(A_\ell^{\circ \ell})$, where the superscript denotes composition, not an ordinary power. Four recent OEIS entries attach a collection of congruence conjectures to the cases $\ell = 3, 4, 5, 6$. We disprove the proposed modulo-5 identity for $\ell = 5$, determine its complete infinite set of exceptions, and prove the other eleven currently conjectural assertions recorded in those entries. The main method gives considerably more: a coefficient modulo p^α is a weighted sum over typed rooted trees on fewer than p^α labels. Consequently every sequence in the family is eventually periodic modulo p^α , with period dividing $p^\alpha(p-1)$ and a uniform onset bound p^α . The ordinary residue generating functions are rational. We also prove periodicity in the iteration parameter modulo the square of the modulus, and classify exactly when $a_\ell(n) \equiv n \pmod{p}$ holds for every n : there is no restriction for $p = 2$, the condition is $3 \mid \ell$ for $p = 3$, and it is $p^2 \mid \ell$ for every prime $p \geq 5$. The first counterexample is checked by two independent exact formal-series algorithms. A reproducible test suite performs 29,122 exact-arithmetic assertions. The proofs, rather than the finite tests, establish the infinite statements.

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1 The selected problem and the outcome

1.1 What was conjectured

The starting point is the following family of formal power series:

$$A_\ell(x) = x \exp(A_\ell^{\circ \ell}(x)), \quad A_\ell(0) = 0, \quad A'_\ell(0) = 1, \quad \ell \in \mathbb{Z}_{\geq 0}. \quad (1)$$

Here $A^{\circ k}$ means k -fold composition, with $A^{\circ 0}(x) = x$. This distinction is essential: replacing composition by $A(x)^\ell$ would give an entirely different problem.

Write

$$A_\ell(x) = \sum_{n \geq 1} a_\ell(n) \frac{x^n}{n!}. \quad (2)$$

The cases $\ell = 3, 4, 5, 6$ are OEIS [A396803](#), [A396804](#), [A396805](#), and [A396806](#), respectively [1, 2, 3, 4]. The entries were introduced in June 2026. In the versions inspected for this report, they contain parity and small-modulus conjectures. The selected assertion is the especially simple claim

$$a_5(n) \equiv n \pmod{5} \quad (n \geq 1). \quad (3)$$

It is attractive because its definition is exact, its truth can be tested without numerical approximation, and any counterexample settles its universal quantifier.

Theorem 1.1 (Complete correction of the selected conjecture). *For every $n \geq 1$,*

$$a_5(n) \equiv \begin{cases} 2, & n = 20j + 3 \text{ or } n = 20j + 4 \text{ for some } j \geq 1, \\ n, & \text{otherwise} \end{cases} \pmod{5}. \quad (4)$$

Thus the first counterexample to (3) is $n = 23$. The exceptional set has natural density $1/10$.

The proof is given in Section 10. It uses only typed trees with at most four vertices, not a long extrapolation from computed terms.

For direct comparison with the OEIS table, the exact value is

$$a_5(23) = 516114659489378430688740267292635767160672734031357. \quad (5)$$

It ends in 7, so its residue modulo 5 is 2, whereas $23 \equiv 3$. The published b-file already contains this integer [5]. The contradiction is therefore already latent in the published data; we do not claim to have discovered a previously unavailable term. The substantive extension is the structural proof, the exact infinite correction, and the general prime-power theory developed below.

1.2 Scope of the resolution

Table 1 records the assertions addressed. Repeated parity assertions are counted separately because they are separately stated in the entries. There are eleven proofs and one disproof among the twelve assertions listed.

A396804 also preserves an older modulo-3 proposal, together with a correction explicitly noting its failure at $n = 5$. That older disproof is credited in the entry to Danial Asaria and is not counted as a new result here. We prove the corrected pattern.

Entry	Assertion in the inspected entry	Outcome here
A396803	Parity agrees with n ; residue modulo 3 agrees with n .	Both proved.
A396804	Parity; $a(n) \equiv n \pmod{4}$; residue 3 never occurs modulo 5; corrected six-term pattern modulo 3.	All four proved.
A396805	Parity; eventual pattern $(0, 1, 0)$ modulo 3 from $n = 3$; $a(n) \equiv n \pmod{5}$.	First two proved; last disproved.
A396806	Parity; agreement with n modulo 3 and modulo 6.	All three proved.

Table 1: The conjectural assertions resolved in this report. Source entries: [1, 2, 3, 4].

1.3 Status and priority

The source checks were performed on September 19, 2026. The assertions in Table 1 remained labelled as conjectures in the inspected entries. No prior resolution of these exact assertions was identified in the checks performed. That is a bounded statement about the sources inspected, not a proof of global publication priority. The use of permutation actions, labelled-tree enumeration, and integer-valued polynomials is standard mathematics; all specialized arguments needed here are supplied in full. The report has not undergone independent peer review or proof-assistant verification.

2 Formal existence and division-free computation

2.1 Existence, uniqueness, and useful identities

Proposition 2.1. *For every nonnegative integer ℓ , equation (1) has a unique solution in $x + x^2\mathbb{Q}[[x]]$.*

Proof. For series with zero constant coefficient and linear coefficient 1, the coefficient of degree d in any fixed compositional iterate depends only on coefficients through degree d of the original series. The coefficient of degree n on the right side of (1) therefore depends only on coefficients of A_ℓ through degree $n - 1$. The linear coefficient is 1. Inductively, the equation determines every subsequent coefficient uniquely, and those recursively determined coefficients satisfy the equation.

Equivalently, start with $A^{[0]} = x$ and repeatedly apply $B \mapsto x \exp(B^{\circ\ell})$. Agreement through degree d improves to agreement through degree $d + 1$. Consequently the coefficients stabilize and give the unique solution. This argument is entirely formal and requires no positive radius of analytic convergence. $\square$

Set $F_k = A_\ell^{\circ k}$ for $k \geq 0$. Substitution into (1) gives

$$F_k = F_{k-1} \exp(F_{\ell+k-1}) \quad (k \geq 1).$$

Multiplying these identities yields

$$F_k = x \exp\left(\sum_{j=\ell}^{\ell+k-1} F_j\right), \quad k \geq 0, \quad (6)$$

where the sum is empty when $k = 0$, and $F_0 = x$. The iteration identities also appear in the OEIS entries; the derivation is included to keep the subsequent interpretation self-contained.

The endpoints of the parameter family are helpful checks:

$$A_0(x) = xe^x, \quad a_0(n) = n; \quad A_1(x) = xe^{A_1(x)}. \quad (7)$$

For $\ell = 1$ the tree interpretation below is the usual enumeration of rooted labelled trees. Lagrange inversion, or Cayley's formula, gives $a_1(n) = n^{n-1}$. That familiar special case is only a check, not an input to any of the congruence proofs.

2.2 An exact recurrence

The standard partial exponential Bell polynomials can be introduced without divisions. Put $B_{0,0} = 1$, $B_{n,0} = 0$ for $n > 0$, and

$$B_{n,k} = \sum_{j=1}^{n-k+1} \binom{n-1}{j-1} a_\ell(j) B_{n-j,k-1} \quad (1 \leq k \leq n). \quad (8)$$

If $c_{k,n} = n![x^n]F_k(x)$, then

$$c_{0,n} = \mathbf{1}_{n=1}, \quad c_{1,n} = a_\ell(n), \quad c_{k,n} = \sum_{j=1}^n c_{k-1,j} B_{n,j} \quad (k \geq 2). \quad (9)$$

The identity follows by expanding a composition into partitions of the n labels: a block of size j receives one of $a_\ell(j)$ inner structures. Formula (8) distinguishes the block containing a particular label.

Let $e_n = n![x^n] \exp(F_\ell(x))$, so $e_0 = 1$. Differentiating the exponential, or selecting the block containing a distinguished label, gives

$$e_n = \sum_{j=1}^n \binom{n-1}{j-1} c_{\ell,j} e_{n-j}, \quad a_\ell(n+1) = (n+1)e_n. \quad (10)$$

These formulas form an incremental algorithm. Once $a_\ell(n)$ is known, compute the n th Bell row, then $c_{k,n}$ in increasing k , then e_n , and finally $a_\ell(n+1)$.

Corollary 2.2. *All $a_\ell(n)$ and all $c_{k,n}$ are nonnegative integers. Moreover, $n \mid a_\ell(n)$ for every $n \geq 1$.*

Proof. All recurrences use only sums, products, and integer binomial coefficients. The last assertion follows immediately from $a_\ell(n) = ne_{n-1}$. $\square$

The first coefficients are

$$\begin{aligned} a_\ell(1) &= 1, & a_\ell(2) &= 2, \\ a_\ell(3) &= 6\ell + 3, & a_\ell(4) &= 48\ell^2 + 12\ell + 4, \\ a_\ell(5) &= 660\ell^3 - 90\ell^2 + 50\ell + 5. \end{aligned} \quad (11)$$

Although these examples are polynomials with integer coefficients, we will not assume that this property persists at every degree. The integer-valued polynomial argument in Section 7 is specifically designed not to make that unjustified inference.

3 A labelled-tree interpretation

Definition 3.1. An ℓ -admissible typed rooted tree is a rooted tree whose vertices carry distinct labels and nonnegative integer types. A vertex of type t may have any unordered set of children, but each child's type s must satisfy

$$\ell \leq s \leq \ell + t - 1. \quad (12)$$

A vertex of type 0 is therefore a leaf. Write $\mathcal{T}_{\ell,k}(n)$ for these trees on a specified set of n labels with root type k . For root type 1, abbreviate $\mathcal{T}_{\ell}(n) = \mathcal{T}_{\ell,1}(n)$.

Labels and types have different roles. Labels identify vertices and are permuted by group actions. Types are integer decorations preserved by those actions. Distinct vertices may have equal types.

For fixed ℓ, k, n , there are finitely many such trees. Along a path, (12) gives $s \leq t + \ell - 1$. For $\ell \geq 1$, every type is at most $k + (n - 1)(\ell - 1)$; for $\ell = 0$, types strictly decrease along edges. Thus all type assignments involved are finite.

Theorem 3.2. *The exponential generating function of $\mathcal{T}_{\ell,k}$ is F_k . In particular,*

$$|\mathcal{T}_{\ell}(n)| = a_{\ell}(n). \quad (13)$$

Proof. Let G_k be the exponential generating function for trees with root type k . Remove the root. Each resulting component is a rooted typed tree whose root type lies in the finite set $\{\ell, \dots, \ell + k - 1\}$. The components form an unordered set on disjoint label subsets. The labelled-set construction gives

$$G_k = x \exp \left(\sum_{j=\ell}^{\ell+k-1} G_j \right).$$

This infinite system has a unique coefficientwise solution: the linear coefficients are 1 and, simultaneously for all k , a coefficient of degree n uses only degrees at most $n - 1$ on the right. Each particular calculation uses only finitely many indices. By (6), F_k solves the same system. Hence $G_k = F_k$. $\square$

Definition 3.3. For a typed tree T , define its total type by

$$S(T) = \sum_{v \in V(T)} t_v.$$

For $r \geq 1$ and $d \geq 0$, define the weighted core count

$$W_{r,d}(\ell) = \sum_{T \in \mathcal{T}_{\ell}(r)} S(T)^d. \quad (14)$$

Throughout, $0^0 = 1$. This convention is necessary when the number of additional orbits is zero. In our root-type-1 family $S(T) \geq 1$, but residue classes of $S(T)$ can be zero.

The interpretation already explains $n \mid a_{\ell}(n)$: every possible root label gives the same count by relabelling. It also turns composition, initially a formal-series operation, into the concrete local rule (12).

4 Reduction modulo a prime: only a small core survives

Theorem 4.1 (Prime-core congruence). *Let p be prime and write $n = pq + r$, where $q \geq 0$ and $0 \leq r < p$. Then*

$$a_{\ell}(n) \equiv \begin{cases} 0, & r = 0, \\ W_{r,q}(\ell), & 1 \leq r < p \end{cases} \pmod{p}. \quad (15)$$

For $n = 0$ set $a_{\ell}(0) = 0$.

Proof. Choose a permutation σ of the n labels consisting of q disjoint p -cycles and r fixed labels. The cyclic group generated by σ acts on $\mathcal{T}_\ell(n)$. Its orbits have sizes 1 or p , so

$$a_\ell(n) \equiv |\text{Fix}(\sigma)| \pmod{p}.$$

A fixed tree must have a fixed root label. Thus there are no fixed trees if $r = 0$.

Assume $r > 0$. The parent of a fixed vertex must itself be fixed: the parent is uniquely determined and the tree is preserved by σ . Consequently the fixed vertices induce a connected rooted tree T on the r fixed labels. They inherit an admissible type assignment, so $T \in \mathcal{T}_\ell(r)$.

Each nonfixed label orbit has size p , and its type is constant. All vertices in such an orbit have parents either in a single nonfixed orbit or at one fixed vertex. Contract the nonfixed orbits. The result is a rooted forest attached to T . No quotient cycle is possible, because iterating the parent map would then prevent reaching the root.

An equivariant parent map from one nonfixed p -orbit to a different nonfixed p -orbit has exactly p choices. Once one vertex's parent is chosen, equivariance determines all the others. These choices are independent for distinct quotient edges. A quotient forest with at least one such edge therefore contributes a multiple of p .

The only remaining case modulo p has every nonfixed orbit directly attached to a fixed vertex. Its p vertices are leaves. An orbit attached to a fixed vertex of type t_v has exactly t_v allowable common leaf types. Summing over the attachment vertex gives $S(T)$ choices for each of the q distinguishable nonfixed orbits. Thus the contribution of T is $S(T)^q$. Summing over T proves the formula. For $q = 0$, this simply counts the core itself. $\square$

The theorem replaces an arbitrarily large coefficient by a finite sum on at most $p - 1$ labels. It is important that it counts fixed trees, not arbitrary partitions of the labels: the parent-map structure is what forces every surviving nonfixed orbit to consist of leaves.

5 The prime-power theorem

A direct repetition of the prime proof is insufficient modulo p^2 : a quotient edge contributes a factor p , not necessarily p^2 . The extra divisibility comes from the number of ways to select the nonfixed orbits participating in a nontrivial quotient forest.

Lemma 5.1 (A forest divisibility estimate). *Fix an admissible core T on $r > 0$ fixed labels and consider a cyclic action of order p on $b > 0$ additional, distinguishable label orbits of size p . The total number of invariant trees with this core and with at least one parent edge between two nonfixed orbits is divisible by*

$$p^{\nu_p(b)+1}.$$

Here “cyclic action of order p ” means that each additional orbit has size p ; the statement includes $p = 2$.

Proof. As in the prime proof, contract the nonfixed orbits to obtain a rooted forest attached to the fixed core. Remove all one-vertex components of this forest. The remaining forest H has D vertices and $e \geq 1$ internal edges. Every component of H has at least two vertices. Keep its types and its attachment vertices in the fixed core as decorations.

For a fixed isomorphism class of decorated H , assigning distinct names from the b available orbits to its vertices can be done in

$$\frac{(b)_D}{|\text{Aut}(H)|}$$

ways. Each internal edge contributes p equivariant parent maps. Each of the remaining $b - D$ orbits is a one-vertex component and has $S(T)$ attachment-and-type choices. The contribution is therefore

$$\frac{(b)_D}{|\text{Aut}(H)|} p^e S(T)^{b-D}. \quad (16)$$

When $b < D$ this class contributes zero and may be omitted.

The action of $\text{Aut}(H)$ on its e directed internal edges is faithful. Indeed, every vertex is an endpoint of such an edge because there are no one-vertex components. An automorphism fixing every directed edge fixes every endpoint, hence every vertex. Thus $\text{Aut}(H)$ embeds into the symmetric group on e objects, and

$$\nu_p(|\text{Aut}(H)|) \leq \nu_p(e!) \leq e - 1.$$

The last inequality follows, for example, from $\nu_p(e!) = (e - s_p(e))/(p - 1)$, where $s_p(e) \geq 1$ is the base- p digit sum; it also follows by counting multiples of $p, p^2, \dots$ directly. Since $D \geq 1$, the falling factorial $(b)_D$ contains the factor b . The p -adic valuation of (16) is at least

$$\nu_p(b) + e - \nu_p(|\text{Aut}(H)|) \geq \nu_p(b) + 1.$$

The quotient in (16) is an integer: it counts labelings of the decorated forest. The valuation calculation is valid even without using this integrality separately. Summing all finitely many classes proves the result. $\square$

Theorem 5.2 (Prime-power core congruence). *Let p be prime, $\alpha \geq 1$, and $m = p^\alpha$. Write $n = mq + r$ with $q \geq 0$ and $0 \leq r < m$. Then*

$$a_\ell(n) \equiv \begin{cases} 0, & r = 0, \\ \sum_{T \in \mathcal{T}_\ell(r)} S(T)^{p^{\alpha-1}q}, & 1 \leq r < m \end{cases} \pmod{m}. \quad (17)$$

The same statement holds for the coefficient sequence of F_k , with cores whose root has type k .

Proof. The case $q = 0$ is just the defining count of the core, so suppose $q > 0$. Choose a permutation σ consisting of q cycles of length p^α and r fixed labels. Let $\tau = \sigma^{p^{\alpha-1}}$, the generator of its order- p subgroup. An orbit of the full cyclic group on the set of trees has size p^j for some $0 \leq j \leq \alpha$. If $j < \alpha$, every point of that orbit is fixed by τ ; if $j = \alpha$, no point is fixed by τ . Therefore

$$|\mathcal{T}_\ell(n)| \equiv |\text{Fix}(\tau)| \pmod{p^\alpha}. \quad (18)$$

On the labels, τ has r fixed points and

$$b = p^{\alpha-1}q$$

nonfixed orbits of size p . If $r = 0$, a fixed root is impossible. Otherwise a τ -fixed tree has a core in $\mathcal{T}_\ell(r)$, exactly as before. By Lemma 5.1, every contribution having a nontrivial quotient component is divisible by $p^{\nu_p(b)+1}$, hence by p^α . The surviving singleton components contribute $S(T)^b$. Combining this count with (18) proves the formula. Nothing in the proof depended on the root type being 1. $\square$

5.1 A wider class of tree specifications

The proof applies more generally when a type t has a finite set of allowable child types, of cardinality $d(t)$. The root type is fixed, children form an unrestricted unordered set, and the rule does not depend on the numerical labels. In this setting replace $S(T)$ by $\sum_v d(t_v)$. The finite-core formula and its prime-power proof remain unchanged. Our family is the special case $d(t) = t$ with allowable set $\{\ell, \dots, \ell + t - 1\}$.

This extension does not include arbitrary degree restrictions, ordered children, or constraints comparing different labels. Such changes can invalidate the independent-leaf-orbit count and require a new argument.

6 Periodicity, rational generating functions, and finite certificates

Corollary 6.1 (Uniform eventual period). *For $m = p^\alpha$,*

$$a_\ell(n + m(p - 1)) \equiv a_\ell(n) \pmod{m} \quad (n \geq m). \quad (19)$$

Thus an eventual period divides $m(p - 1)$, uniformly in ℓ .

Proof. For $n = mq + r$ with $q \geq 1$, the exponent in (17) is $p^{\alpha-1}q$. If $p \nmid S$, Euler's theorem gives

$$S^{p^{\alpha-1}(q+p-1)} \equiv S^{p^{\alpha-1}q} \pmod{p^\alpha}.$$

If $p \mid S$, both sides vanish: the exponent is at least $p^{\alpha-1} \geq \alpha$. The elementary inequality $2^{\alpha-1} \geq \alpha$ proves the latter bound for all $p \geq 2$. The case $r = 0$ is zero on both sides. $\square$

Corollary 6.2 (Rational residue generating function). *Let $R_{\ell,m}(z) = \sum_{n \geq 1} (a_\ell(n) \bmod m) z^n$ be regarded as a formal series over $\mathbb{Z}/m\mathbb{Z}$. Then*

$$R_{\ell,m}(z) = \sum_{r=1}^{m-1} z^r \sum_{T \in \mathcal{T}_\ell(r)} \frac{1}{1 - S(T)p^{\alpha-1}z^m}. \quad (20)$$

Every denominator has constant coefficient 1, so the expression is well defined even though $\mathbb{Z}/m\mathbb{Z}$ need not be a field.

Proof. For each r and T , sum (17) over $q \geq 0$ as a geometric series. Only finitely many cores occur. $\square$

If $S(T)$ is divisible by p , then $S(T)p^{\alpha-1} = 0$ modulo m and its summand is simply z^r . Otherwise the same base has order dividing $p - 1$. Thus a common denominator is $1 - z^{m(p-1)}$, with a finite polynomial accounting for the initial coefficients. In particular, multiplying $R_{\ell,m}$ by that denominator produces a polynomial of degree at most $m(p - 1) + m - 1$.

For a general modulus $M = \prod_i p_i^{\alpha_i}$, the Chinese remainder theorem gives the explicit eventual period bound

$$L(M) = \text{lcm}_i(p_i^{\alpha_i}(p_i - 1)), \quad n \geq \max_i p_i^{\alpha_i}. \quad (21)$$

An actual least period may be smaller. The bound is an infinite theorem, not a pattern guessed from a finite residue table.

6.1 What constitutes a finite certificate?

For fixed ℓ and $m = p^\alpha$, it is enough to record, for each $1 \leq r < m$, the distribution of $S(T)$ modulo m over $\mathcal{T}_\ell(r)$. Evaluating a coefficient then consists of a finite weighted power sum. A certificate can be large for a large prime power, but its size does not depend on the queried index n .

Alternatively, after the theorem is established, one may compute the first $m(p - 1) + m - 1$ coefficients by the division-free recurrence and reduce all later indices using (19). For fixed modulus this gives random access to a residue using a finite table and modular reduction of the index. The large integer $a_\ell(n)$ never needs to be constructed.

7 Periodicity in the iteration parameter and an exact classification

7.1 Weighted cores are integer-valued polynomials

Lemma 7.1. *For $r \geq 2$ and $d \geq 0$, $W_{r,d}(\ell)$ is a polynomial in ℓ with rational coefficients, takes integer values at all nonnegative integers, and has degree at most $d + r - 2$. Moreover,*

$$W_{r,d}(0) = r. \quad (22)$$

For $r = 1$, $W_{1,d}(\ell) = 1$.

Proof. Fix an underlying rooted labelled tree on r vertices. Every child of the root has forced type ℓ , since the root has type 1. If the root has $c \geq 1$ children, there are $r - 1 - c \leq r - 2$ remaining variable types. The summand S^d has total degree d in those types and ℓ .

Sum variable types from the leaves toward the root. A variable of type s ranges from ℓ to $\ell + t - 1$, where t is its parent's type. Summing a polynomial over such an interval produces a rational polynomial and raises total degree by at most one. This follows by expressing powers in a binomial-polynomial basis and using $\sum_{j=0}^{N-1} \binom{j}{k} = \binom{N}{k+1}$; equivalently it is polynomial discrete antidifferentiation. After at most $r - 2$ sums the degree is at most $d + r - 2$. Summing the finitely many underlying trees preserves this bound. Integrality at nonnegative ℓ follows from the original count.

When $\ell = 0$, the root's children all have type 0 and hence are leaves. Only rooted stars are possible, one for each of the r root labels. Each has total type 1, proving (22). $\square$

We need integer-valued, not necessarily integer-coefficient, polynomials. Every polynomial P of degree at most D taking integer values on nonnegative integers has a Newton expansion

$$P(X) = \sum_{j=0}^D c_j \binom{X}{j}, \quad c_j = \Delta^j P(0) \in \mathbb{Z}. \quad (23)$$

The expansion follows from finite differences and agrees with P at all nonnegative integers, hence as a polynomial.

Lemma 7.2. *Let $m = p^\alpha$. An integer-valued polynomial of degree $D < 2m$ satisfies*

$$P(L + m^2) \equiv P(L) \pmod{m} \quad (L \geq 0).$$

Proof. For $1 \leq i \leq D$, we have $i < 2p^\alpha \leq p^{\alpha+1}$, so $v_p(i) \leq \alpha$. Since

$$\binom{m^2}{i} = \frac{m^2}{i} \binom{m^2 - 1}{i - 1},$$

its valuation is at least $2\alpha - v_p(i) \geq \alpha$. Vandermonde's identity now gives

$$\binom{L + m^2}{j} = \sum_{i=0}^j \binom{m^2}{i} \binom{L}{j-i} \equiv \binom{L}{j} \pmod{m}$$

for every $j \leq D$. Apply (23). $\square$

Theorem 7.3 (Parameter periodicity). *For every integer $M \geq 2$, every $\ell \geq 0$, and every $n \geq 1$,*

$$a_{\ell+M^2}(n) \equiv a_\ell(n) \pmod{M}. \quad (24)$$

In particular,

$$M^2 \mid \ell \implies a_\ell(n) \equiv n \pmod{M} \text{ for all } n \geq 1. \quad (25)$$

Proof. First take $M = m = p^\alpha$. Write $n = mq + r$. If $r = 0$, both coefficients are zero modulo m by divisibility by n . If $r = 1$, the core is a single vertex of total type 1, so the residue is independent of ℓ .

For $2 \leq r < m$, the core exponent is zero when $q = 0$. When $q > 0$, reduce q to its representative $t \in \{1, \dots, p-1\}$ modulo $p-1$. As in the proof of Corollary 6.1, units have the same power and nonunits vanish, so the residue is $W_{r,d}(\ell)$ with $d = p^{\alpha-1}t \leq \varphi(m)$. In either case Lemma 7.1 gives degree at most

$$d + r - 2 \leq \varphi(m) + m - 3 < 2m.$$

Lemma 7.2 proves the claimed parameter period m^2 . For arbitrary M , apply the prime-power result to each factor $p^\alpha \parallel M$. The shift M^2 is a multiple of each $p^{2\alpha}$, so the Chinese remainder theorem proves (24). Finally $a_0(n) = n$, and repeated parameter shifts give (25). $\square$

7.2 Which parameters give the identity modulo a prime?

Theorem 7.4 (Complete prime-modulus classification). *Fix a prime p and a nonnegative integer ℓ . Then*

$$a_\ell(n) \equiv n \pmod{p} \text{ for every } n \geq 1$$

holds exactly under the following conditions:

p	necessary and sufficient condition
2	no restriction on ℓ
3	$3 \mid \ell$
$p \geq 5$	$p^2 \mid \ell$.

(26)

Proof. The cases 2 and 3 follow from the explicit formulas in Section 9. For $p \geq 5$, the value $a_\ell(3) = 6\ell + 3$ forces $p \mid \ell$; write $\ell = ph$. On three labels the core is a star or a path. Section 8 gives

$$W_{3,q}(\ell) = 3(1 + 2\ell)^q + 6 \sum_{t=\ell}^{2\ell-1} (1 + \ell + t)^q.$$

Modulo p , the sum for $\ell = ph$ contains h full sets of residues. For $q = p-1$, it is $-h$ because the nonzero residues have $(p-1)$ st power 1 and zero contributes zero. The prime-core theorem, with $n = p(p-1) + 3$, therefore gives

$$a_{ph}(p(p-1) + 3) \equiv 3 - 6h \pmod{p}. \quad (27)$$

Since $p \geq 5$, agreement with $n \equiv 3$ forces $p \mid h$. Thus $p^2 \mid \ell$ is necessary. It is sufficient by Theorem 7.3. $\square$

In fact the same calculation gives an infinite family of witnesses:

$$a_{ph}(p(p-1)j + 3) \equiv 3 - 6h \pmod{p} \quad (p \geq 5, j \geq 1). \quad (28)$$

For $p \nmid h$, every one of these indices violates agreement with n . We do not claim that this progression always begins at the first failure for every prime; only the case $p = 5$ receives a complete exception-set analysis below.

8 Small cores and explicit formulas

The following finite sums are the computational heart of the concrete applications. They are exact integer identities, before any modular reduction. Let $I_\ell = \{\ell, \dots, 2\ell - 1\}$; it is empty for $\ell = 0$.

$$W_{1,d}(\ell) = 1, \quad (29)$$

$$W_{2,d}(\ell) = 2(1 + \ell)^d, \quad (30)$$

$$W_{3,d}(\ell) = 3(1 + 2\ell)^d + 6 \sum_{t \in I_\ell} (1 + \ell + t)^d. \quad (31)$$

For four vertices,

$$\begin{aligned} W_{4,d}(\ell) &= 4(1 + 3\ell)^d + 24 \sum_{t \in I_\ell} (1 + 2\ell + t)^d \\ &\quad + 12 \sum_{s, t \in I_\ell} (1 + \ell + s + t)^d \\ &\quad + 24 \sum_{t \in I_\ell} \sum_{u=\ell}^{\ell+t-1} (1 + \ell + t + u)^d. \end{aligned} \quad (32)$$

Here is a direct derivation, including the labelled multiplicities. A two-vertex tree is a path with two choices for its root label; its child has type ℓ . On three vertices there are three rooted stars, with types $(1, \ell, \ell)$, and six rooted paths, with types $(1, \ell, t)$ for $t \in I_\ell$.

On four vertices the four rooted shapes are as follows. A star has four choices of root and types $(1, \ell, \ell, \ell)$. A root with two children, one of which has one child, has $4 \cdot 3 \cdot 2 = 24$ labelled realizations and types $(1, \ell, \ell, t)$. A root with one child that has two leaves has $4 \cdot 3 = 12$ realizations and types $(1, \ell, s, t)$. A path has $4! = 24$ realizations and types $(1, \ell, t, u)$, with $t \in I_\ell$ and $\ell \leq u \leq \ell + t - 1$. These are precisely the four terms in (32).

At $d = 0$ the formulas give $1, 2, 6\ell + 3, 48\ell^2 + 12\ell + 4$, agreeing with (11). Independent Prüfer-code enumeration in the supplied program checks both the multiplicities and the complete total-type distributions; the displayed derivation establishes them without code.

9 Complete formulas modulo 2, 3, 4

9.1 Parity for the entire family

Corollary 9.1. *For every $\ell \geq 0$ and $n \geq 1$,*

$$a_\ell(n) \equiv n \pmod{2}.$$

Proof. If n is even, divisibility by n suffices. If $n = 2q + 1$, the prime-core formula has a one-vertex core with total type 1, giving $a_\ell(n) \equiv 1^q = 1$. $\square$

9.2 The full modulo-3 formula

Corollary 9.2. *For $q \geq 0$,*

$$a_\ell(3q) \equiv 0, \quad a_\ell(3q + 1) \equiv 1, \quad a_\ell(3q + 2) \equiv 2(1 + \ell)^q \pmod{3}. \quad (33)$$

Proof. Use (29)–(30) in Theorem 4.1 for $p = 3$. □

When $\ell \equiv 0 \pmod{3}$, the residues agree with n . When $\ell \equiv 1 \pmod{3}$, the pattern from $n = 1$ is

$$1, 2, 0, 1, 1, 0, \quad \text{repeating.}$$

When $\ell \equiv 2 \pmod{3}$, the values at $n = 1, 2$ are 1, 2, and from $n = 3$ onward the pattern is 0, 1, 0 repeating. These three cases prove the relevant assertions for all four OEIS entries. They also show the necessity of $3 \mid \ell$ for agreement with n at every index: take $q = 1$ in the third formula.

9.3 The full modulo-4 formula

Corollary 9.3. *For every $\ell \geq 0$ and $q \geq 0$,*

$$\begin{aligned} a_\ell(4q) &\equiv 0, & a_\ell(4q+1) &\equiv 1, \\ a_\ell(4q+2) &\equiv 2(1+\ell)^{2q} \pmod{4}. \end{aligned} \tag{34}$$

For the remaining residue class,

$$a_\ell(3) \equiv 3 + 2\ell \pmod{4}, \quad a_\ell(4q+3) \equiv 3 + 2\left\lceil \frac{\ell}{2} \right\rceil \pmod{4} \quad (q \geq 1). \tag{35}$$

In particular, $a_\ell(n) \equiv n \pmod{4}$ for all $n \geq 1$ if and only if $4 \mid \ell$.

Proof. Apply Theorem 5.2 with $p = 2$, $\alpha = 2$. The exponent is $2q$, so (29) and (30) give (34). For a three-vertex core, the star term in (31) is 3 because $1 + 2\ell$ is odd. For $q \geq 1$, an even integer to the power $2q$ is zero modulo 4, and an odd integer to that power is one. The quantity $1 + \ell + t$ is odd precisely when t has the same parity as ℓ . Exactly $\lceil \ell/2 \rceil$ integers in I_ℓ do. Since $6 \equiv 2 \pmod{4}$, formula (35) follows. Its $q = 0$ case is the exact value $6\ell + 3$.

If agreement with n holds for every n , the index 3 forces ℓ to be even, and the index 7 then forces $\ell/2$ to be even. Conversely, $4 \mid \ell$ makes every displayed formula agree with n . □

The case $\ell = 4$ proves the modulo-4 conjecture for A396804. Combining parity with the modulo-3 result proves agreement modulo 6 whenever $3 \mid \ell$, including $\ell = 6$ as conjectured in A396806.

10 The modulo-5 conjecture: all exceptions and a larger family

We prove more than Theorem 1.1 by allowing any parameter $\ell = 5h$.

Lemma 10.1. *For $q \geq 0$, put*

$$U_q = \sum_{s=0}^4 s^q \pmod{5}, \quad \delta_q = \mathbf{1}_{q>0 \text{ and } 4 \mid q}.$$

Then $U_q = -\delta_q$ modulo 5.

Proof. For $q = 0$, the convention $0^0 = 1$ gives $U_0 = 5 = 0$ modulo 5. For $q > 0$, the zero term vanishes. The nonzero residues form a cyclic group of order 4; their q th powers sum to zero unless $4 \mid q$, in which case the sum is $4 = -1$. For this small prime one can also check the four exponent residues $q = 1, 2, 3, 4$ directly and use $s^4 = 1$ for $s \neq 0$. □

Theorem 10.2 (Exact modulo-5 formula for parameters divisible by 5). *Let $h \geq 0$. For every $n \geq 1$,*

$$a_{5h}(n) \equiv \begin{cases} 3 - h, & n = 20j + 3, j \geq 1, \\ 4 - 2h, & n = 20j + 4, j \geq 1, \\ n, & \text{otherwise} \end{cases} \pmod{5}. \quad (36)$$

Proof. Write $n = 5q + r$. The cases $r = 0, 1, 2$ follow at once from the prime-core formula: the residues are respectively 0, 1, 2 because $1 + 5h \equiv 1$. For $r = 3$, formula (31) consists of a star contribution 3 and a sum over $5h$ consecutive type values. They contain h copies of each residue modulo 5. Therefore

$$a_{5h}(5q + 3) \equiv 3 + 6hU_q \equiv 3 + hU_q = 3 - h\delta_q \pmod{5}. \quad (37)$$

For $r = 4$, use the four terms in (32). The star contributes 4. The root-with-two-children term contributes $24hU_q$. The double sum with multiplicity 12 contributes $12 \cdot 5h^2U_q = 0$ modulo 5. It remains to evaluate the path term.

Write $t = 5h + 5j + s$, where $0 \leq j < h$ and $0 \leq s < 5$. The inner interval for u has $h + j$ complete residue blocks, followed by residues $0, \dots, s - 1$. Because the total type is congruent to $1 + s + u$, the inner sum equals

$$(h + j)U_q + \sum_{v=0}^{s-1} (1 + s + v)^q \pmod{5}.$$

After summing over s , the complete-block part is a multiple of 5. Thus the entire double sum is

$$hV_q, \quad V_q = \sum_{0 \leq v < s \leq 4} (1 + s + v)^q.$$

The summand is symmetric in s, v . The sum over all ordered pairs is $5U_q = 0$, and the diagonal sum is U_q because $s \mapsto 1 + 2s$ permutes the residues. Hence

$$2V_q = -U_q, \quad V_q = -\frac{1}{2}U_q \pmod{5}.$$

Combining the four shapes gives

$$a_{5h}(5q + 4) \equiv 4 + 24hU_q - 12hU_q = 4 + 12hU_q = 4 - 2h\delta_q \pmod{5}. \quad (38)$$

Equations (37)–(38) are exactly (36), since $\delta_q = 1$ precisely for $q = 4j$ with $j \geq 1$. $\square$

Taking $h = 1$ proves Theorem 1.1. The first exceptional indices are

$$23, 24, 43, 44, 63, 64, 83, 84, 103, 104, \dots$$

The terms $a_5(1), \dots, a_5(22)$ really do agree with their indices modulo 5; the theorem explains why a short initial check is misleading.

10.1 A rational generating-function correction

Let $R_{5h,5}(z) = \sum_{n \geq 1} a_{5h}(n)z^n$ over $\mathbb{F}_5$. Since $\sum_{n \geq 1} nz^n = z/(1 - z)^2$, the exact residue formula is equivalent to

$$R_{5h,5}(z) = \frac{z}{(1 - z)^2} - h \frac{z^{23} + 2z^{24}}{1 - z^{20}} \quad \text{in } \mathbb{F}_5[[z]]. \quad (39)$$

This is an ordinary generating function for the residues, not the original exponential generating function A_{5h} .

For $h = 1$, the least eventual period is exactly 20, and this 20-periodicity starts at $n = 5$. To see minimality, observe that in the periodic tail the zero residues occur exactly at multiples of 5. Every period is therefore divisible by 5. Since 20 is a period, a least period must be one of 5, 10, 20: the least period of a periodic word divides every period, as follows by Euclidean division of shifts. Periods 5 and 10 both fail, comparing $n = 23$ with $n = 28$ and $n = 33$ and translating these comparisons by arbitrarily large multiples of 20. Finally, the comparisons $a_5(3) \neq a_5(23)$ and $a_5(4) \neq a_5(24)$ prevent an onset before 5.

There are exactly two exceptional residue classes in every block of length 20 after the initial block. Their natural density is therefore $2/20 = 1/10$. In a full periodic block, residues 0, 1, 2, 3, 4 occur with frequencies 4, 4, 6, 3, 3, respectively.

10.2 An immediate repaired conjecture

Within the subfamily $\ell = 5h$, agreement with n modulo 5 at every index holds exactly when $5 \mid h$, or equivalently $25 \mid \ell$. The general prime classification proves that no other parameters work. Thus replacing the proposed parameter 5 by a multiple of 25 repairs the identity; the original modulus and its strikingly simple right side can be retained.

11 The missing residue for A396804

The neighboring conjecture that $a_4(n)$ is never 3 modulo 5 is true. We give an explicit certificate for every index.

For $1 \leq r \leq 4$, let $N_r(s)$ be the number of cores in $\mathcal{T}_4(r)$ with $S(T) \equiv s \pmod{5}$. Equations (29)–(32), or their shape-by-shape derivation, give the exact counts in Table 2. For example, the row $r = 3$ has three

r	$N_r(0)$	$N_r(1)$	$N_r(2)$	$N_r(3)$	$N_r(4)$	Total
1	0	1	0	0	0	1
2	2	0	0	0	0	2
3	6	6	6	0	9	27
4	180	192	132	160	156	820

Table 2: Exact total-type frequencies for parameter $\ell = 4$.

stars of total type 9, and six paths for each $t = 4, 5, 6, 7$, with total type $5 + t$. For $r = 4$, the following four contribution vectors provide a compact manual certificate for the less immediate last row:

shape	$s = 0$	$s = 1$	$s = 2$	$s = 3$	$s = 4$
star	0	0	0	4	0
root with two children	24	24	0	24	24
fork below root	36	48	36	36	36
path	120	120	96	96	96

(40)

Each entry follows by substituting $t, s \in \{4, 5, 6, 7\}$ in the corresponding term of (32). Their column sums give Table 2.

By the prime-core theorem,

$$a_4(5q + r) \equiv \sum_{s=0}^4 N_r(s) s^q \pmod{5}.$$

For $q > 0$, the answer depends only on $q \bmod 4$. The resulting complete table is

	$q = 0$	$q \equiv 1$	$q \equiv 2$	$q \equiv 3$	$q \equiv 0 \pmod{4}, q > 0$
$r = 0$	0	0	0	0	0
$r = 1$	1	1	1	1	1
$r = 2$	2	0	0	0	0
$r = 3$	2	4	4	0	1
$r = 4$	0	0	1	2	0

(41)

No entry equals 3.

Corollary 11.1. *For all $n \geq 1$, $a_4(n) \not\equiv 3 \pmod{5}$.*

This proves the residue-avoidance conjecture without assuming any of the other conjectures in A396804.

For comparison, the exact core frequencies for $\ell = 5$ are

r	$s = 0$	$s = 1$	$s = 2$	$s = 3$	$s = 4$
1	0	1	0	0	0
2	0	2	0	0	0
3	6	9	6	6	6
4	252	256	252	252	252.

(42)

Modulo 5, these yield $W_{3,q} = 3 + U_q$ and $W_{4,q} = 4 + 2U_q$, providing a second small certificate for the corrected modulo-5 formula in the case $h = 1$.

12 Verification, algorithms, and reproducibility

12.1 What the computation establishes

The supplied test suite completed **29,122 exact-arithmetic assertions**. Its broad modular grid covers $0 \leq \ell \leq 12$ and $1 \leq n \leq 260$. The checks include the following distinct mechanisms:

1. The coefficient $a_5(23)$ and all earlier coefficients agree between an incremental Bell-polynomial recurrence and an independent ordinary power-series implementation using exact rational arithmetic.
2. For $0 \leq \ell \leq 8$ and core sizes at most four, the explicit shape formulas agree with independent labelled-tree enumeration using Prüfer codes and direct enumeration of allowable types.
3. The parity, modulo-3, modulo-4, and modulo-5 formulas agree with direct coefficient calculations on their stated finite test grids. Prime-power core checks include moduli 4, 8, 9, 25, in addition to primes.
4. Eventual-period checks and parameter-square-period checks are made against direct calculations of the original functional equation. The coefficient routine does not first reduce the parameter or the index by the theorem it is intended to test.

The exact counterexample is not inferred from a rounded approximation or from its magnitude. Both independent computations produce precisely the integer in (5). The original OEIS b-file supplies an external third comparison for that value.

The finite checks do not prove the infinite statements. For example, testing periodicity through 260 cannot exclude a later failure. That exclusion is supplied by the cyclic-group and forest-divisibility proofs. Conversely, the independent implementations help detect transcription, indexing, or multiplicity errors in the presentation.

12.2 Algorithmic details

The main coefficient routine implements (8)–(10). For a truncation through N , the operation count is

$$O(N^3 + \ell N^2),$$

and the storage is $O(N^2 + \ell N)$. These are arithmetic-operation bounds, not bit-complexity bounds: unrestricted integer coefficients grow rapidly. For a fixed modulus, reduction after each sum keeps all stored coefficients bounded by that modulus.

The second routine stores ordinary coefficients as Python `Fraction` objects. It uses ordinary polynomial convolution, Horner composition, and the recurrence

$$[x^n]e^{H(x)} = \frac{1}{n} \sum_{j=1}^n j[x^j]H(x) [x^{n-j}]e^{H(x)}.$$

It then iterates the original functional equation until the requested truncation has stabilized. This is slower but avoids sharing the Bell polynomial implementation with the main computation.

The direct core enumerator decodes every Prüfer word of length $r - 2$, chooses every possible root, orients the resulting labelled tree, and assigns types recursively from the root. It is deliberately restricted to $r \leq 5$ in the public interface, because its purpose is independent verification of small certificates rather than large-scale enumeration.

12.3 Pitfalls avoided

An exponential generating function should not be reduced modulo a prime by treating $x^n/n!$ as an ordinary field coefficient when $p \mid n!$. Our modular algorithm instead reduces the integer sequences and uses only integer polynomial recurrences.

Similarly, using only a prime-order action would not justify the prime-power statement. The passage from the full cyclic group to its order- p subgroup, followed by the valuation estimate on decorated forests, is essential. Finally, the cases $q = 0$ must be kept separate when powers of zero residues occur. Confusing 0^0 with 0 would erase the initial exceptions responsible for the first false conjecture.

12.4 Archive contents and commands

The archive is self-contained apart from a standard Python installation and a LaTeX distribution. No third-party Python packages are required.

```
python3 code/verify.py
python3 code/iterative_series.py 5 40
```

```
python3 code/iterative_series.py 5 260 --modulus 5
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The data/ directory contains exact early coefficients, the modular grid, core-frequency certificates, general-prime witnesses, and a JSON verification summary. The README specifies the scope of each file. Running the verification program regenerates these artifacts. No checksum files, font files, or external literature PDFs are bundled.

13 Further questions and conclusion

The selected conjecture has a complete negative answer, and all eleven other live assertions in Table 1 have proofs. Several natural refinements remain worthwhile.

The prime-power period bound is uniform, but it need not be the least period. Determining the exact eventual period and the shortest initial segment for each pair (ℓ, p^α) is a finite problem after the core theorem, and a more conceptual classification may be possible. The parameter-square period is also a bound rather than a universal minimal period. Modulo 2 the parameter does not matter at all; modulo 3 its period is only 3.

Theorem 7.4 completely classifies agreement with n modulo a prime. A comparably sharp classification modulo arbitrary prime powers would strengthen the sufficient condition $p^{2\alpha} \mid \ell$. Our proof gives a finite decision procedure for each fixed modulus but does not claim that this sufficient condition is necessary.

A separate direction is the real or complex asymptotic growth of the integer coefficients as $n \rightarrow \infty$ for fixed $\ell \geq 2$. No analytic convergence or asymptotic expansion is asserted in this report. The arithmetic results are formal and combinatorial, and their validity does not depend on resolving that analytic question.

The central lesson is specific rather than merely methodological. For this iterative exponential family, congruences of arbitrarily large coefficients are controlled by a fixed finite collection of small typed trees. A numerical contradiction at $n = 23$ becomes an exact infinite exception set, rational residue generating functions, and a uniform prime-power theory.

A A compact proof of the main counterexample family

For readers wishing to verify only the selected conjecture's resolution, the following is a short route through the full proof. The typed-tree interpretation identifies $a_5(n)$ with rooted labelled trees having root type 1 and child types in $\{5, \dots, t+4\}$ when the parent has type t . Apply a permutation with q disjoint 5-cycles and r fixed labels, where $n = 5q + r$ and $0 \leq r < 5$. Modulo 5, invariant trees with an edge between two nonfixed label orbits vanish, because such an edge has five possible equivariant parent maps. Every surviving orbit consists of leaves attached directly to a fixed-label core. Thus its contribution is the core's total type to the q th power.

The total-type distributions on one through four fixed labels are the four rows of (42). Reducing those rows modulo 5 gives

$$\begin{array}{c|ccccc} 1 & 0 & 1 & 0 & 0 & 0 \\ 2 & 0 & 2 & 0 & 0 & 0 \\ 3 & 1 & 4 & 1 & 1 & 1 \\ 4 & 2 & 1 & 2 & 2 & 2. \end{array}$$

Consequently the residue at $n = 5q + r$ is $0, 1, 2, 3 + U_q, 4 + 2U_q$ for $r = 0, 1, 2, 3, 4$, respectively. Since

$U_q = 0$ except when q is a positive multiple of 4, and $U_q = -1$ in that case, the only failures are $n = 20j + 3$ and $20j + 4$ with $j \geq 1$. At both kinds of index the residue is 2. The first index is 23.

Every count in this argument is supported by an explicit small-tree classification. Its infinite quantifier comes from the group action, not from the size of a computational search.

B Selected exact data and modular witnesses

The first twelve terms for $\ell = 5$ are

n	$a_5(n)$	$a_5(n) \bmod 5$
1	1	1
2	2	2
3	33	3
4	1264	4
5	80505	0
6	7365456	1
7	893006317	2
8	136393773888	3
9	25351203412449	4
10	5592284787086560	0
11	1436624546038113861	1
12	423481344402695225472	2

These initial values agree with the source entry and its b-file [3, 5]; they are independently regenerated in the archive. The file `a396805_first40.csv` extends the exact list through $n = 40$.

The general witness formula (27) gives:

p	ℓ	$n = p(p-1) + 3$	$a_\ell(n) \bmod p$	$n \bmod p$
5	5	23	2	3
7	7	45	4	3
11	11	113	8	3
13	13	159	10	3

The table is finite corroboration of an already proved formula. Only for $p = 5$ does this report assert that the displayed index is the first failure and list every exceptional index.

References

- [1] Paul D. Hanna and contributors, *OEIS A396803: E.g.f. satisfies $A(x) = x \exp(A^{\circ 3}(x))$* , created June 8, 2026; inspected September 19, 2026. <https://oeis.org/A396803>.
- [2] Paul D. Hanna, Danial Asaria, Seiichi Manyama, and contributors, *OEIS A396804: E.g.f. satisfies $A(x) = x \exp(A^{\circ 4}(x))$* , created June 9, 2026; inspected September 19, 2026. <https://oeis.org/A396804>. The entry records the correction to the earlier modulo-3 proposal.
- [3] Paul D. Hanna, Seiichi Manyama, and contributors, *OEIS A396805: E.g.f. satisfies $A(x) = x \exp(A^{\circ 5}(x))$* , June 2026; inspected September 19, 2026. <https://oeis.org/A396805>.
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- [5] Paul D. Hanna, *Table of $n, a(n)$ for OEIS A396805, $1 \leq n \leq 200$* , inspected September 19, 2026. <https://oeis.org/A396805/b396805.txt>. In particular, the entry at index 23 already contradicts the printed modulo-5 conjecture.